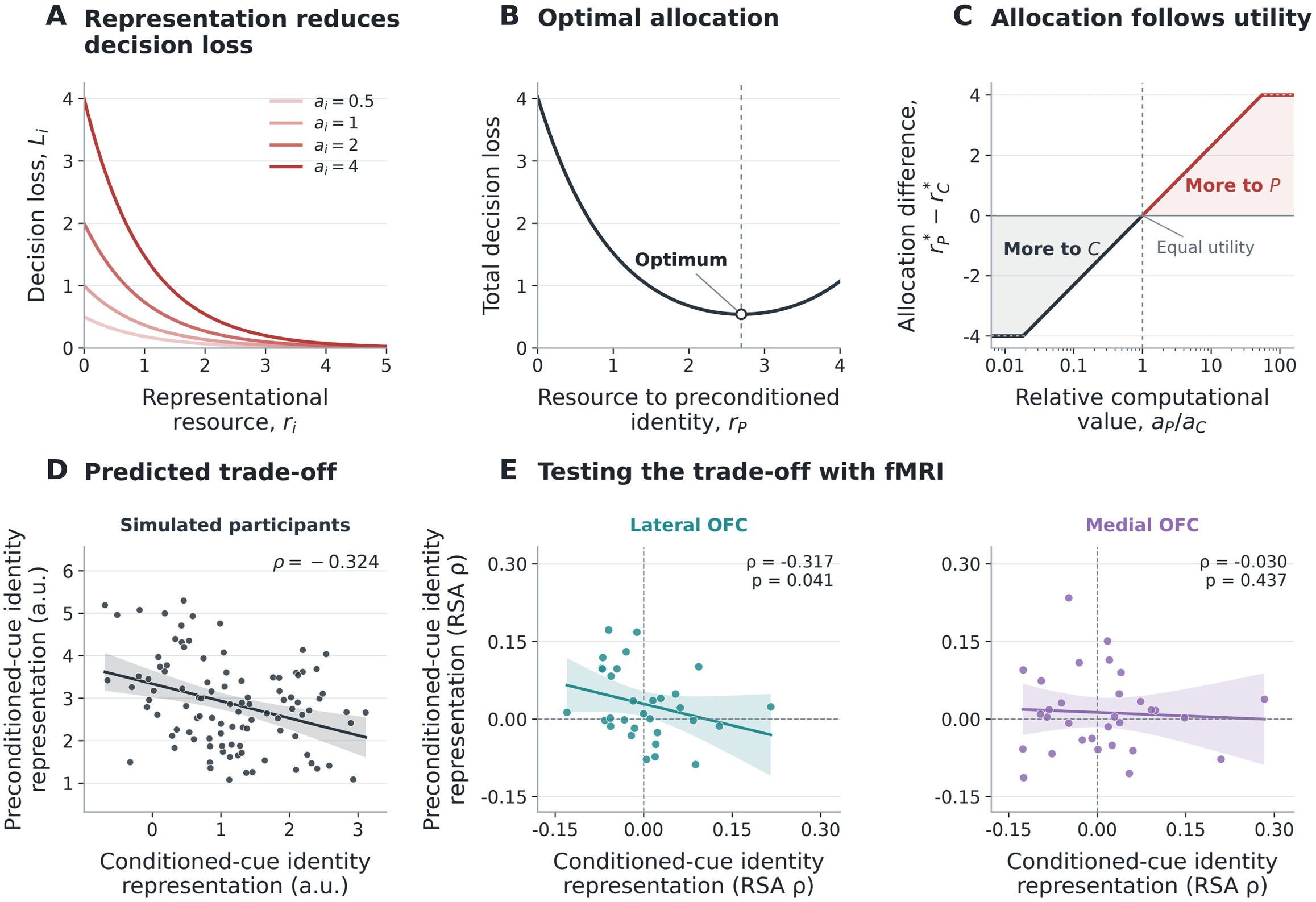

A Representation reduces decision loss
Decision loss, L_i
$a_i = 0.5$
$a_i = 1$
$a_i = 2$
$a_i = 4$
Representational resource, r_i
B Optimal allocation
Total decision loss
Optimum
Resource to preconditioned identity, r_P
C Allocation follows utility
Allocation difference, $r_P^* - r_C^*$
More to P
More to C
Equal utility
Relative computational value, a_P/a_C
D Predicted trade-off
Simulated participants
$\rho = -0.324$
Preconditioned-cue identity representation (a.u.)
Conditioned-cue identity representation (a.u.)
E Testing the trade-off with fMRI
Lateral OFC
ρ = -0.317
p = 0.041
Preconditioned-cue identity representation (RSA ρ)
Conditioned-cue identity representation (RSA ρ)
Medial OFC
ρ = -0.030
p = 0.437
Conditioned-cue identity representation (RSA ρ)